

Positive radial series and sharp logarithmic concavity on balls

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September 15, 2026

Abstract

We study positive Dirichlet solutions of $-\Delta u = u \log u^2$ on a ball in any dimension. We prove that $-\log(u(r)/u(0))$ has a power series in r^2 with strictly positive coefficients and convergence radius equal to the square of the ball radius. The coefficients follow an explicit positive recurrence and are polynomials with nonnegative coefficients in the excess of the central logarithmic height above the Gaussian threshold. This structure proves concavity of $-\sqrt{-\log(u/\|u\|_\infty)}$, answering a question of Gallo, Mosconi and Squassina. The exponent $1/2$ is sharp among positive powers of the negative normalized logarithm. We also obtain a quantitative lower bound for its Hessian before taking the square root, positive power concavity of u on every ball, and a comparison of normalized profiles. The argument gives an alternative proof of the known shooting order and ball uniqueness theorem. The essential step is to pass from a local positive recurrence to a convergent representation throughout the domain.

Mathematics Subject Classification (2020). 35B45, 35J61, 34B15, 26A51.

Keywords. Logarithmic Schrödinger equation; concavity; radial solutions; positive power series; shooting method.

1. Introduction and main results

Let $N \geq 1$, $R > 0$, and $B_R = \{x \in \mathbb{R}^N : |x| < R\}$. We consider

$$-\Delta u = u \log u^2 \quad \text{in } B_R, \quad u > 0 \quad \text{in } B_R, \quad u = 0 \quad \text{on } \partial B_R, \quad (1.1)$$

where $u \in C^2(B_R) \cap C(\overline{B_R})$. All concavity statements about a logarithmic transform concern the open ball. Statements about u^α extend to the closed ball by continuity.

Gallo, Mosconi and Squassina [GMS] establish existence of a solution with locally negative-definite logarithmic Hessian on every bounded convex domain. Their Theorem 6.5 gives radial symmetry and radial decrease of every positive solution on a ball. In Sections 6.2--6.3 they ask for stronger concavity in balls: positive power concavity of u , and concavity of the negative square root of its negative normalized logarithm. Their Theorem 6.6 proves both properties in dimension one. The second question remains explicitly stated on page 44 of the final article.

Our main result is a positive-series representation that answers both questions in every dimension. Write $u(x) = U(r)$, $r = |x|$, and set

$$M = \log U(0), \quad w(r) = -\log \frac{U(r)}{U(0)}, \quad A = \frac{2M}{N}. \quad (1.2)$$

Theorem 1.1 (Positive radial series). Every solution of (1.1) satisfies $A > 1$. There are strictly positive coefficients c_n , $n \geq 0$, such that

$$\frac{w'(r)}{r} = \sum_{n=0}^{\infty} c_n r^{2n}, \quad w(r) = \sum_{j=1}^{\infty} \frac{c_{j-1}}{2j} r^{2j}, \quad 0 \leq r < R. \quad (1.3)$$

The quotient at $r = 0$ is interpreted by continuity. Both series, regarded as series in r^2 , have radius of convergence exactly R^2 . Their coefficients are determined by

$$c_0 = A, \quad c_{n+1} = \frac{(n+1) \sum_{i=0}^n c_i c_{n-i} - c_n}{(n+1)(2n+N+2)}. \quad (1.4)$$

In particular, $Z(t) = w'(\sqrt{t})/\sqrt{t}$ is absolutely monotone on $[0, R^2)$: every derivative is strictly positive there. The same is true of every derivative of positive order of $W(t) = w(\sqrt{t})$. In addition,

$$D^2 w(|x|) \succeq A I_N, \quad (1.5)$$

with strict inequality as quadratic forms when $x \neq 0$. Here $\succeq$ denotes the order on symmetric matrices.

Theorem 1.2 (Sharp transformed concavity). For every solution of (1.1),

$$x \mapsto - \left(-\log \frac{u(x)}{\|u\|_{\infty}} \right)^{\beta} \text{ is concave on } B_R \iff \beta \geq \frac{1}{2}, \quad \beta > 0. \quad (1.6)$$

For $\beta = 1/2$ its Hessian is negative definite away from the center. At the center it is continuous and has a norm-type cusp; no differentiability assertion there is intended.

Theorem 1.3 (Positive power concavity). There exists a number $\alpha_*(U) \in (0, 1)$ such that, for $\alpha > 0$,

$$u^{\alpha} \text{ is concave on } \overline{B_R} \iff \alpha \leq \alpha_*(U). \quad (1.7)$$

It is characterized by an attained interior minimum,

$$\alpha_*(U) = \min_{0 < r < R} \frac{w''(r)}{w'(r)^2}. \quad (1.8)$$

The formula is implicit. We do not assert a positive exponent uniform over all radii, or monotonicity of this optimal exponent as a function of the radius.

1.1. Prior results and the scope of the contribution

The amplitude threshold, uniqueness in balls, and strict ordering of first zeros are prior results. For $k = 1$ and $\beta = 0$, Ben Chrouda [BC, Theorem 1.2 and Corollaries 1.3 and 1.5] proves these facts for $-\Delta v = v \log v$. The change $v(y) = u(y/\sqrt{2})$ gives the normalization (1.1). The equality case in the

amplitude threshold is excluded by the Gaussian solution and uniqueness of the central initial-value problem. We give new proofs of these particular consequences because they fit the recurrence argument, but do not claim their first discovery.

Liu, Sun and Zou [LSZ] prove a broader shooting classification of whole-space nodal solutions; in particular, their argument establishes strict motion of zeros with the initial value. Gallo, Moraschi and Squassina [GMoS] develop parabolic concavity principles whose logarithmic example yields preservation of logarithmic concavity. Neither of these results supplies the positive radial series used here. Our priority claim concerns that structure and the resulting stronger concavity, rather than existence, radial symmetry, Gaussian ground-state classification, or shooting uniqueness. Current versions of these sources were compared on September 15, 2026.

The proof of Theorem 1.1 has three stages. An integrating factor puts the logarithmic radial derivative above its Gaussian value. Its equation then produces positive Taylor coefficients. Finally, positivity and a real-axis singularity argument show that the series represents the solution all the way to its first zero. Convexity of a Hilbert-space norm proves Theorem 1.2. A separate boundary argument, which accommodates the singular derivative of $u \log u^2$ at zero, completes Theorem 1.3.

2. The radial equation and the Gaussian threshold

We use radial symmetry and radial decrease from [GMS, Theorem 6.5]. These are nontrivial inputs: the derivative of the nonlinearity is unbounded below near zero, so an unqualified appeal to the usual locally Lipschitz moving-plane theorem would be inappropriate. Existence follows from [GMS, Theorem 1.1], with its one-dimensional result covering $N = 1$. The rest of the argument applies to every positive radial decreasing solution.

Put $f(s) = s \log s^2$ for $s > 0$, and $f(0) = 0$. The radial equation and its integral form are

$$U'' + \frac{N-1}{r}U' + f(U) = 0, \quad U'(r) = -r^{1-N} \int_0^r s^{N-1} f(U(s)) ds. \quad (2.1)$$

The central initial-value problem is unique as long as its solution stays positive. Indeed, integrating (2.1) once more gives a Volterra equation. On a sufficiently small interval the Lipschitz bound for f near a prescribed positive central value makes its integral operator a contraction, since the norm of the double integral is bounded by a constant times r^2 . Ordinary uniqueness then continues the solution away from zero. The same identity, using smoothness of f near a positive central value, gives the usual even expansions at the center.

Since U has its maximum at zero, (2.1) implies $f(U(0)) \geq 0$, hence $M \geq 0$. If $M = 0$, central uniqueness would give $U \equiv 1$, which violates the boundary condition. Thus $M > 0$.

In fact $U'(r) < 0$ for $0 < r < R$. If U' vanished there, its nonpositivity would imply $U'' = 0$ at that point. Equation (2.1) would give $U = 1$, and ordinary uniqueness with data $(U, U') = (1, 0)$ would again force the constant solution.

In terms of (1.2), equation (2.1) becomes

$$w'' + \frac{N-1}{r}w' - (w')^2 = 2M - 2w. \quad (2.2)$$

Define $z(r) = w'(r)/r$ for $r > 0$ and extend it to the center. The central expansion gives

$$z(0) = \frac{2M}{N} = A, \quad z'(r) = O(r). \quad (2.3)$$

Differentiation of (2.2) yields

$$z'' + \left(\frac{N+1}{r} - 2rz \right) z' = 2z(z-1). \quad (2.4)$$

Equivalently,

$$\left(r^{N+1} e^{-2w} z' \right)' = 2r^{N+1} e^{-2w} z(z-1). \quad (2.5)$$

Lemma 2.1. We have $A > 1$ and $z'(r) > 0$ for all $0 < r < R$.

Proof. Suppose first that $0 < A < 1$. Integrating (2.5) from zero shows that $z' < 0$ while $0 < z < 1$. The boundary term at zero vanishes by (2.3). The function cannot first reach 1, since it is decreasing. It cannot first reach zero either: the integral in (2.5) would then give a strictly negative derivative, whereas $z = w'/r > 0$ throughout the interior. Thus $0 < z \leq A$ on $[0, R)$, giving $w(r) \leq Ar^2/2$. This contradicts $w(r) \rightarrow \infty$ as $r \uparrow R$.

If $A = 1$, the central solution is

$$U(r) = \exp \left(\frac{N}{2} - \frac{r^2}{2} \right), \quad (2.6)$$

by the uniqueness following (2.1). It has no finite zero. Hence $A > 1$. Now (2.5) gives $z' > 0$ as long as $z > 1$, and this strict increase prevents any first exit from that region. $\square$

The eigenvalues of $D^2 w(|x|)$ are $w'' = z + rz'$ in the radial direction and z in the tangential directions. Lemma 2.1 proves (1.5), including its strict form away from the center. In dimension one only the radial eigenvalue is present.

3. A positive recurrence valid throughout the ball

Let $t = r^2$ and $Z(t) = z(\sqrt{t})$. Equation (2.4) becomes

$$2tZ'' + (N+2)Z' - 2tZZ' = Z(Z-1). \quad (3.1)$$

Lemma 3.1 (Local construction). For each $A > 1$, recurrence (1.4) defines strictly positive coefficients satisfying

$$0 < c_n \leq A^{n+1}, \quad n \geq 0. \quad (3.2)$$

The resulting series solves (3.1) near zero and agrees with the transformed radial solution.

Proof. The coefficient of t^n in the linear derivative terms of (3.1) is $(n+1)(2n+N+2)c_{n+1}$. In the nonlinear terms, symmetry of convolution gives

$$[t^n](Z^2 + 2tZZ') = (n+1) \sum_{i=0}^n c_i c_{n-i}, \quad (3.3)$$

which proves (1.4). In particular $c_1 = A(A-1)/(N+2) > 0$. For $n \geq 1$, its numerator can be written

$$(2(n+1)A-1)c_n + (n+1) \sum_{i=1}^{n-1} c_i c_{n-i}, \quad (3.4)$$

so positivity follows by induction. Dropping the negative term in (1.4) and using the inductive upper bounds gives

$$c_{n+1} \leq \frac{(n+1)A^{n+2}}{2n+N+2} \leq A^{n+2}. \quad (3.5)$$

Consequently the series converges for $|t| < 1/A$, and termwise differentiation verifies (3.1).

Integrate $w'(r) = rZ(r^2)$ with $w(0) = 0$. The derivative of the difference between the two sides of (2.2) vanishes by (3.1). Its limiting value at zero is $NA - 2M = 0$. Thus (2.2) holds. Setting $U(r) = e^{M-w(r)}$ recovers the radial equation with the prescribed central data. The uniqueness argument after (2.1) identifies this constructed solution with the given one. $\square$

We next state the positive-series continuation principle, including its short proof.

Lemma 3.2 (Positive real singularity). Suppose $F(t) = \sum_{n \geq 0} a_n t^n$ has $a_n \geq 0$ and finite positive radius of convergence ρ . It cannot admit a holomorphic continuation to a neighborhood of the positive real point ρ .

Proof. If such a continuation existed, continuity of its derivatives and monotone convergence as $t \uparrow \rho$ would give, for every $k \geq 0$,

$$F^{(k)}(\rho) = \sum_{n \geq k} \frac{n!}{(n-k)!} a_n \rho^{n-k}. \quad (3.6)$$

Choose a sufficiently small $h > 0$ in the convergence disk of the Taylor series centered at ρ . Its coefficients are nonnegative by (3.6). Interchanging nonnegative sums gives

$$\begin{aligned} \sum_{k \geq 0} \frac{F^{(k)}(\rho)}{k!} h^k &= \sum_{n \geq 0} a_n \sum_{k=0}^n \binom{n}{k} \rho^{n-k} h^k \\ &= \sum_{n \geq 0} a_n (\rho + h)^n < \infty, \end{aligned} \quad (3.7)$$

contradicting the definition of ρ . $\square$

Proof of Theorem 1.1. By Lemma 3.1 the series for Z represents the actual solution near zero. Denote its convergence radius by ρ . If $\rho < R^2$, the actual solution is regular at the positive point ρ . On an interval around that point, (3.1) is an ordinary differential equation with analytic right-hand side after solving for Z'' . It therefore has a locally holomorphic solution with the actual values of Z and Z' there. Ordinary uniqueness identifies this solution with the continuation from zero. This contradicts Lemma 3.2. Thus $\rho \geq R^2$.

Conversely, $z(r)$ tends to infinity as $r \uparrow R$: by Lemma 2.1 it is increasing, and a finite limit would make $w(r) = \int_0^r sz(s) ds$ bounded. Therefore $\rho > R^2$ is impossible. We have proved $\rho = R^2$. Integration gives the second series in (1.3). Dividing its coefficients by a factor linear in the index leaves the convergence radius unchanged. All derivative assertions now follow by differentiating convergent positive series on compact subintervals of $[0, R^2)$. The Hessian assertion was proved after Lemma 2.1. $\square$

The continuation step is essential: positivity of finitely many coefficients, or even a formal positive recurrence without convergence control, would not imply a global concavity theorem.

4. Convexity of the square root and sharpness

Put $a_j = c_{j-1}/(2j) > 0$. Theorem 1.1 gives

$$\sqrt{w(r)} = \left\| (\sqrt{a_j} r^j)_{j \geq 1} \right\|_{\ell^2}, \quad 0 \leq r < R. \quad (4.1)$$

Every coordinate on the right is nonnegative, nondecreasing and convex as a function of r . Coordinatewise convexity, monotonicity of the ℓ^2 norm on nonnegative sequences, and the triangle inequality show that $r \mapsto \sqrt{w(r)}$ is convex and nondecreasing. Equivalently one may first prove this for the finite partial sums and pass to the finite pointwise limit on $[0, R)$.

For strictness it is useful to give an exact identity. At $r > 0$ let

$$p_j = \frac{a_j r^{2j}}{w(r)}, \quad m = \sum_{j \geq 1} j p_j, \quad v = \sum_{j \geq 1} (j - m)^2 p_j. \quad (4.2)$$

These are a probability distribution, its mean and its variance; all moments are finite inside the convergence interval. Direct differentiation gives

$$\frac{r^2(2ww'' - (w')^2)}{4w^2} = 2v + m(m - 1) > 0. \quad (4.3)$$

The strict inequality holds because $m > 1$, as $a_2 > 0$. Thus $(\sqrt{w})'' > 0$ for $r > 0$. Its tangential Hessian eigenvalue is $(\sqrt{w})'/r > 0$.

Proof of Theorem 1.2. A convex nondecreasing function of the Euclidean norm is convex. Applying this fact to (4.1) proves convexity of $x \mapsto \sqrt{w(|x|)}$ across the center as well. The Hessian assertions away from the center follow from (4.3). For $\beta \geq 1/2$, the function $s \mapsto s^{2\beta}$ is convex and nondecreasing on $[0, \infty)$; composing it with $\sqrt{w}$ proves the forward implication of (1.6).

For necessity, the central expansion is

$$w(r) = \frac{A}{2} r^2 + O(r^4). \quad (4.4)$$

If $0 < \beta < 1/2$, the restriction $w(r)^\beta$ to a positive radial segment has strictly negative second derivative for sufficiently small $r > 0$. Indeed its leading term is $(A/2)^\beta r^{2\beta}$, whose second derivative has the negative factor $2\beta(2\beta - 1)$; the differentiated remainder has higher order. It cannot be convex. Finally, (4.4) gives $\sqrt{w(|x|)} = \sqrt{A/2}|x| + O(|x|^3)$, which describes the stated cusp. $\square$

The same series gives more than a Hessian bound. For every integer $k \geq 1$, the radial differential operator $\mathcal{D} = (2r)^{-1}d/dr$ satisfies $\mathcal{D}^k w(r) > 0$ on $0 < r < R$, with a positive limit at zero. These infinitely many sign inequalities refer to differentiation with respect to squared radius, not arbitrary mixed spatial derivatives.

5. Boundary behavior and the optimal positive power

Because f is bounded on the compact range of U , (2.1) gives a finite limit of U' at R . The differential equation then gives a continuous limit of U'' there. The next argument rules out a zero boundary slope without assuming a finite value of $f'(0)$.

Lemma 5.1 (Simple boundary zero). There is $b > 0$ such that

$$U(r) = b(R - r) + O((R - r)^2), \quad U'(r) = -b + O(R - r), \quad (5.1)$$

and

$$z(r) \sim \frac{1}{R(R - r)} \quad \text{as } r \uparrow R. \quad (5.2)$$

Proof. Write $v = -U' > 0$. Near R we have $0 < U < 1$, and (2.1) gives $U'' > 0$. Thus v is decreasing there. Set

$$F(s) = \frac{s^2}{2}(\log s^2 - 1), \quad E(r) = \frac{v(r)^2}{2} + F(U(r)). \quad (5.3)$$

The energy identity is $E' = -(N - 1)v^2/r$. Suppose $U'(R) = 0$. Then $E(R) = 0$, and for r sufficiently close to R ,

$$E(r) = \int_r^R \frac{N - 1}{s} v(s)^2 ds \leq C(R - r)v(r)^2. \quad (5.4)$$

Choose this neighborhood so that $C(R - r) \leq 1/4$. Equation (5.4) implies

$$\frac{v(r)^2}{4} \leq -F(U(r)), \quad v(r) \leq C_1 U(r) \sqrt{\log(1/U(r))}. \quad (5.5)$$

Since U is strictly decreasing, it can be used as the variable of integration on this interval. The remaining time to reach zero would satisfy

$$R - r = \int_0^{U(r)} \frac{ds}{v(U^{-1}(s))} \geq \frac{1}{C_1} \int_0^{U(r)} \frac{ds}{s \sqrt{\log(1/s)}} = \infty, \quad (5.6)$$

a contradiction. When $N = 1$, the integral in (5.4) is zero and the same argument applies. Thus $b = -U'(R) > 0$. The bounded second derivative gives (5.1), and $z = -U'/(rU)$ gives (5.2). $\square$

Proof of Theorem 1.3. For $r > 0$,

$$(U^\alpha)'' = \alpha U^\alpha (\alpha (w')^2 - w''), \quad \frac{(U^\alpha)'}{r} = -\alpha U^\alpha \frac{w'}{r} < 0. \quad (5.7)$$

Thus the radial Hessian condition is precisely $\alpha \leq q(r)$, where $q = w''/(w')^2$. The condition at the center holds automatically, because $D^2 u^\alpha(0) = -\alpha U(0)^\alpha A I_N$. By Lemma 2.1, $q > 0$. Equation (4.4) gives $q(r) \rightarrow \infty$ at zero, while Lemma 5.1 gives

$$q(r) = 1 - \frac{U(r)U''(r)}{U'(r)^2} \longrightarrow 1 \quad \text{as } r \uparrow R. \quad (5.8)$$

Moreover $U'' > 0$ near R , so $q < 1$ there. Choose any such point. The limits at both endpoints and continuity imply that the global infimum is achieved on a compact interior interval; its value is strictly between zero and one. Equations (5.7) prove (1.7)–(1.8), and continuity extends concavity to the closed ball. $\square$

6. Coefficient comparison and two consequences

The recurrence contains information about variation with the central height. Fix N and put $\delta = A - 1 > 0$. Write $c_n(\delta)$ for its coefficients.

Proposition 6.1 (Positive parameter dependence). For every $n \geq 1$, $c_n(\delta)$ is a polynomial in δ with nonnegative coefficients and zero constant term. If two solutions have $0 < \delta_1 < \delta_2$ and $\theta = \delta_2/\delta_1$, then on their common radial interval,

$$z_2(r) - 1 \geq \theta(z_1(r) - 1), \quad (6.1)$$

and

$$w_2(r) - \frac{r^2}{2} \geq \theta \left(w_1(r) - \frac{r^2}{2} \right). \quad (6.2)$$

Proof. We have $c_1 = (\delta + \delta^2)/(N + 2)$. For $n \geq 1$, rewrite (1.4) as

$$c_{n+1} = \frac{(2n + 1 + 2(n + 1)\delta)c_n + (n + 1) \sum_{i=1}^{n-1} c_i c_{n-i}}{(n + 1)(2n + N + 2)}. \quad (6.3)$$

The polynomial assertion follows by induction. Every monomial of positive degree satisfies $\delta_2^k \geq \theta \delta_1^k$. Apply this coefficientwise, including the constant-in- r term $c_0 - 1 = \delta$, and sum the convergent series to obtain (6.1). Integrating $r(z_i(r) - 1)$ gives (6.2). $\square$

Corollary 6.2 (Known shooting order and uniqueness). If $U_2(0) > U_1(0)$, their first zero radii satisfy $R_2 < R_1$. Consequently (1.1) has exactly one positive solution for every $R > 0$.

Proof. Suppose $R_2 \geq R_1$. By (6.1) and divergence of z_1 at R_1 , strict inequality $R_2 > R_1$ is impossible. If $R_2 = R_1$, (5.2) gives $z_2/z_1 \rightarrow 1$, contradicting (6.1) and $\theta > 1$. This proves the strict shooting order. Two distinct positive solutions on the same ball are radial and have either different central values, contradicted by this order, or equal central values, contradicted by initial-value uniqueness. Existence is the prior input stated in Section 2. $\square$

As emphasized in Section 1.1, this is an alternative proof of a prior theorem. The coefficientwise parameter comparison (6.1)–(6.2) is the additional conclusion of the present method.

Corollary 6.3 (Products of balls). Let u_i be positive solutions of (1.1) on $B_{R_i} \subset \mathbb{R}^{N_i}$, for $1 \leq i \leq k$. Then

$$u(x_1, \dots, x_k) = \prod_{i=1}^k u_i(x_i) \quad (6.4)$$

solves the same equation on the product domain, and $-\sqrt{-\log(u/\|u\|_\infty)}$ is concave there.

Proof. Dividing the Laplacian of the product by the product gives $-\Delta u/u = \sum_i \log u_i^2 = \log u^2$. The negative normalized logarithm is $\sum_i w_i$. Its square root is the Euclidean norm of the vector $(\sqrt{w_i})_{i=1}^k$. Each component is nonnegative and convex in its corresponding variables by Theorem 1.2. The argument of (4.1) proves convexity of this norm. $\square$

This includes the interval-product construction already treated in [GMS]; the factorization itself is elementary and is not a separate novelty claim.

7. Further questions

The argument uses the squared-radius equation and does not extend the square-root-log conclusion to arbitrary convex domains. Whether that conclusion survives controlled deformations of a ball, and what geometric obstructions can prevent it, remain natural questions. The positivity proof also depends on the particular coefficient structure in (3.1); a classification of nonlinearities or radial potentials admitting a comparable positive recurrence would be useful.

Formula (1.8) does not evaluate the optimal positive concavity exponent. Determining its dependence on the radius in dimensions $N \geq 2$, especially its sharp limiting behavior, is a further problem. These questions and the related consequences above belong to the same research direction; no result about them is asserted here.

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Preparation note

This manuscript was prepared with OpenAI Codex assistance in literature searches, proof development and writing. It is a research preprint. No independent human verification or journal acceptance is claimed.